

Practice Problems

1. Show that if a prime p divides a^n then $p^n \mid a^n$.
2. Suppose a, n are positive integers and $a^{1/n}$ is rational. Then show that $a^{1/n}$ must be an integer. Use this to show $10^{1/3}$ is not rational.
3. Suppose that m and n are positive integers such that $\gcd(m, n) = 1$. If d is positive and $d \mid mn$, then there are unique positive integers d_1 and d_2 such that $d = d_1 d_2$, $d_1 \mid m$, $d_2 \mid n$.
4. Characterize when $ax + by = c$ will have a solution for integers x and y , where a, b, c are given integers. In case a solution exists, give an algorithm to find a solution.
5. Show that a number is divisible by 1001 iff the alternating sum of its 3-digit blocks is divisible by 1001. For example $n = 4176204105$ is not divisible by 1001 because $105 - 204 + 176 - 4 = 73$ is not divisible by 1001.
6. Number of steps in Euclid's Algorithm to find $\gcd(a, b)$: Let $a < b$. If Euclid's Algorithm takes N division steps to find the $\gcd(a, b)$, then show that $a \geq F_{N+1}$, where $F_0, F_1, \dots$ are Fibonacci numbers given by $F_0 = 0, F_1 = 1$ and $F_n = F_{n-1} + F_{n-2}$. Thus conclude that the number of division steps needed by the Euclidean algorithm to find the greatest common divisor of two positive integers does not exceed five times the number of digits (in decimal, i.e. base 10) in the smaller of the two integers.

Also show that if a, b are consecutive Fibonacci numbers, say F_{k+1} and F_{k+2} where $k > 1$, then the number of division steps is k . This means that the upper bound on the number of steps (as shown above) is tight.
7. Let $n_1, n_2, \dots, n_r$ be positive integers and $b_1, b_2, \dots, b_r$ be any integers. Consider the system of congruences

$$\begin{cases} x \equiv b_1 \pmod{n_1}, \\ x \equiv b_2 \pmod{n_2}, \\ \vdots \\ x \equiv b_r \pmod{n_r}. \end{cases}$$

Show that

- (a) The system has a solution if and only if

$$\gcd(n_i, n_j) \mid (b_i - b_j) \quad \text{for all } i, j.$$

- (b) Whenever a solution exists, it is unique modulo

$$\text{lcm}(n_1, n_2, \dots, n_r).$$

8. Solve for x

- (a) $3x \equiv 2 \pmod{78}$
- (b) $6x \equiv 10 \pmod{14}$
- (c) $9x \equiv 21 \pmod{12}$
- (d) $x \equiv 2 \pmod{3}, x \equiv 3 \pmod{4}$
- (e) $x \equiv 7 \pmod{9}, x \equiv 13 \pmod{23}, x \equiv 1 \pmod{2}$
- (f) $2x \equiv 3 \pmod{5}, 4x \equiv 3 \pmod{7}$

9. If p is a prime then $(p-1)! \equiv -1 \pmod{p}$.

10. Find the remainder when

- (a) 3^{31} is divided by 7,
- (b) 128^{129} is divided by 17,
- (c) 29^{25} is divided by 11.

11. Solve the congruence $x^{103} \equiv 4 \pmod{11}$.

12. Suppose p, q are distinct primes such that $a^p \equiv a \pmod{q}$ and $a^q \equiv a \pmod{p}$. Then $a^{pq} \equiv a \pmod{pq}$.

13. Show $\prod_{p \text{ prime}, p \leq n} p \leq 4^n$.

(Hint: first show $\binom{2n+1}{n} \geq \prod_{p \in \{n+2, n+3, \dots, 2n+1\}, p \text{ prime}} p$
and do induction on n .)

14. Show that for large n ,

$$\frac{n}{\log_2 n} - 2 < \pi(n) < \frac{6n}{\log_2 n}$$

where $\pi(n)$ is the number of primes in $\{1, 2, \dots, n\}$.

(Hint: use the result in Q. 13 to show the upper bound on $\pi(n)$. For the lower bound, use the result shown in class — if for prime p , p^ℓ divides $\binom{2n}{n}$ then $p^\ell \leq 2n$, and that $\binom{2n}{n} \geq 4^n/(2n+1)$.)

Below are some interesting problems in Graph theory that I missed in the last assignment.

1. A vertex cover of a graph G is a subset of vertices $S \subseteq V(G)$ that touches every edges of the graph, i.e., any edge $e \in E(G)$ will have at least one of its endpoints in S . Obviously, $V(G)$ is a vertex cover. We are interested in vertex cover of minimum size.

Show that for *bipartite graphs*, the size of maximum matching is equal to the size of minimum vertex cover (i.e., vertex cover of minimum size)

2. The diameter of a set of points in the plane is the maximum distance (measured in the usual Euclidean distance) between two points of the set. Let S be a finite set of points of diameter one in the plane. Then the number of pairs of points in S whose distance is greater than $1/\sqrt{2}$ is at most $n^2/3$.

Group Theory

1. Given an integer $n > 0$, let

$$G = \left\{ e^{\frac{2\pi i k}{n}} : k \in \mathbb{Z} \right\}.$$

Show that G is a group under multiplication. How many elements does G have?

2. Let $(G, *)$ be a group in which the square of every element is the identity. Show that G is abelian.
3. Prove that every group of order less or equal to five is abelian.
4. If S and T are subgroups of group G then show that $S \cap T$ is also a subgroup. If S is a subgroup of G show that there exists a smallest subgroup of G (under inclusion) containing S .
5. We have seen that for any $a \in G$, $\langle a \rangle = \{\dots, a^{-2}, a^{-1}, 1, a, a^2, \dots\}$ is the subgroup generated by a . We can generalize this definition to any subset $S \subseteq G$. The subgroup generated by S is defined as set all elements that can be expressed as a finite product of elements in S and its inverses. Show that this subgroup is the smallest subgroup containing S (as defined in the previous question).
6. Product formula: If X, Y are subgroups of a finite group G . Let XY be defined as

$$XY = \{xy : x \in X, y \in Y\}$$

then show that $|X||Y| = |X \cap Y||XY|$

7. Show that every subgroup of a cyclic group is cyclic.
8. Let G be a cyclic group of order n generated by a . Then G has $\phi(n)$ generators where ϕ is Euler totient function.

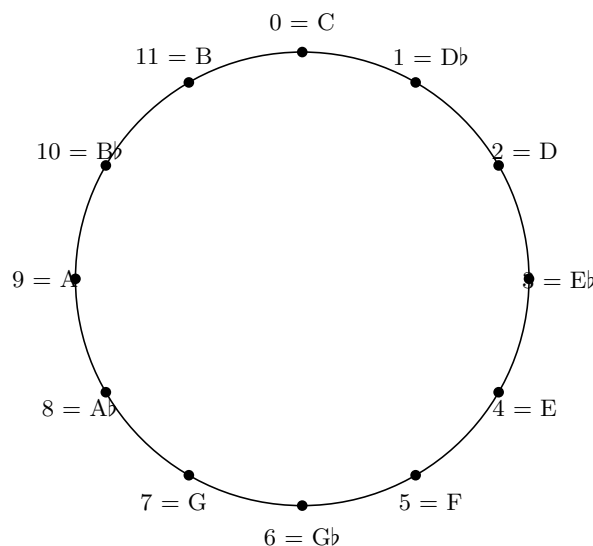
9. Let H be a subgroup of S_n (symmetric group). Show that either all the permutations in H are even or that exactly half of them are.
10. If G is a group of even order, prove it has an element of order 2.
11. We say element x is conjugate to element y in a group G if there exists $g \in G$ such that $y = g^{-1}xg$. Show that this is an equivalence relation. Show that
 - (a) In group D_3 , the conjugate equivalence classes are $\{\{1\}, \{r, r^2\}, \{s_1, s_2, s_3\}\}$.
 - (b) In group D_4 , the conjugate equivalence classes are $\{\{1\}, \{r, r^3\}, \{r^2\}, \{s_1, s_2\}, \{s_3, s_4\}\}$ where s_1, s_2 are reflections passing through opposite vertices and s_3, s_4 are reflections passing through mid point of opposite sides.
 - (c) In a symmetric group, two permutations π and σ are conjugate iff for each $k \geq 1$, the number of cycles of length k in both π and σ are same.

From above examples, conclude that the two group elements are conjugate if they have similar structure.

12. Let G be a group of order $2n$ and let $r, s \in G$ such that $r^n = 1, s^2 = 1, sr^{-1} = rs$. Show that group G is isomorphic to the dihedral group D_n .
13. You have 4 red, 4 white, and 4 blue identical dinner plates. In how many different ways can you set a square table with one plate on each side if two settings are different only if you cannot rotate the table to make the settings match?
14. Show that there are total 34 non-isomorphic simple graphs on 5 vertices.
15. How many distinct coloring are possible for a necklace with 7 beads with n colors if two necklaces are considered equivalent upto rotations and reflections.
16. In Western music, tones are equivalent if they differ by an octave. There are a total of 12 tones, which are

C, D $\flat$, D, E $\flat$, E, F, G $\flat$, G, A $\flat$, A, B $\flat$, B.

The tones are typically arranged in a circle to show that they cycle through an octave. They can be represented by the numbers 0 through 11, as shown below.



A *triad* (chord) is any subset of three distinct tones. Two triads are considered *equivalent* if one can be obtained from the other by a *translation* or by a *permutation* (reordering of the same notes).

For example,

$$[C, E, G] = [0, 4, 7] \quad \text{and} \quad [F, A, C] = [5, 9, 0]$$

are equivalent because each note in the second chord is 5 semitones higher than in the first. Formally, $[a, b, c]$ and $[d, e, f]$ are equivalent if there exists k such that $[a + k, b + k, c + k] = [d, e, f]$ where addition is modulo 12. Similarly, $[C, E, G]$ and $[E, G, C]$ are equivalent as they are same upto ordering.

How many distinct triads are possible under this equivalence relation?

17. For $n \geq 2$, show that $n - 1$ transpositions $(1, 2), (1, 3), \dots, (1, n)$ generate S_n (recall the definition of generation defined in Q 5).

Fields

1. Determine whether the following polynomials are irreducible:
 - (a) $x^2 + 1$ over $\mathbb{F}_2$,
 - (b) $x^2 + 1$ over $\mathbb{F}_3$,
 - (c) $x^2 - 2$ over $\mathbb{Q}$ (rational field).
 - (d) $x^2 - 2$ over $\mathbb{R}$.
2. **Constructing a Field of Size 2^3 .** Find a monic irreducible polynomial of degree 3 over the field $\mathbb{F}_2 = \{0, 1\}$. Using this polynomial, construct a field of size $2^3 = 8$. (Write out both the addition table and the multiplication table.)

3. Let

$$a(x) = 2x^3 + 2x^2 + 2 \in \mathbb{F}_3[x].$$

Compute $-a(x)$, $2a(x)$, and $a(x)^2$, writing all coefficients in $\mathbb{F}_3$.

4. Find the greatest common divisor of the polynomials $x^2 + 1$ and $x^3 + 1$ over each of the following fields $\mathbb{F}_2, \mathbb{R}$.

5. Let $f(x)$ and $g(x)$ be polynomials with real coefficients. Since $\mathbb{R} \subseteq \mathbb{C}$, these polynomials also belong to $\mathbb{C}[x]$. Let $h(x)$ denote a greatest common divisor of $f(x)$ and $g(x)$ in $\mathbb{C}[x]$. Show that $h(x)$ has real coefficients. Thus, $h(x)$ is also the gcd of f and g in $\mathbb{R}[x]$.

6. Let $\mathbb{F}$ be a field. For each of the following pairs of polynomials $f(x), g(x) \in \mathbb{F}[x]$, compute $\gcd(f, g)$ and find polynomials $u(x), v(x) \in \mathbb{F}[x]$ such that

$$u(x)f(x) + v(x)g(x) = \gcd(f, g).$$

(a) Over $\mathbb{R}$:

$$f(x) = x^3 - 2x, \quad g(x) = x^2 - 1.$$

(b) Over $\mathbb{F}_2$:

$$f(x) = x^4 + x + 1, \quad g(x) = x^3 + 1.$$

7. Let $\mathbb{F}$ be any field. For an integer $k \geq 0$ and $a \in \mathbb{F}$, let ka denote the sum $a + a + \cdots + a$ (k times), with $0a = 0$.

(a) Will the Binomial Theorem hold in $\mathbb{F}$?, i.e., for all $a, b \in \mathbb{F}$

$$(a + b)^n = \sum_{r=0}^n \binom{n}{r} a^r b^{n-r}.$$

(b) If a finite $\mathbb{F}$ has characteristic p (recall this is always a prime), show that for any $a, b \in \mathbb{F}$,

$$(a + b)^p = a^p + b^p.$$

8. **The Multiplicative Group of a Finite Field is always Cyclic.** Let $\mathbb{F}$ be a finite field. You will show that its multiplicative group $(\mathbb{F} \setminus \{0\}, \cdot)$ is cyclic.

(a) Structure of any finite cyclic group: Let $G = \langle a \rangle$ be a finite cyclic group of order n . Recall that G has exactly $\phi(n)$ generators, and every subgroup of G is cyclic (Questions 7,8 in Group theory). Also, by Lagrange theorem, the size of any subgroup must divide n . Prove that for each divisor d of n , there is exactly one cyclic subgroup of G of order d .

(b) Show that

$$n = \sum_{d|n} \phi(d).$$

(You may use the result from part (a).)

- (c) Recall the fact proved in class: in the multiplicative group of any field, there can be *at most one* cyclic subgroup of any given order.
- (d) Let $|\mathbb{F}| = q$. Then $|\mathbb{F} \setminus \{0\}| = q - 1$. For the sake of contradiction, assume that every element of $\mathbb{F} \setminus \{0\}$ has order strictly less than $q - 1$. Obtain a contradiction from the following identity, derived from part (b)

$$q - 1 = \sum_{d|(q-1)} \phi(d).$$

Note: A generator of this multiplicative group is called primitive element.